

Calculus A: Lecture 4

Vectors, products, and geometry in 3D space

Carlos Pérez Arancibia
(c.a.perezarancibia@utwente.nl)

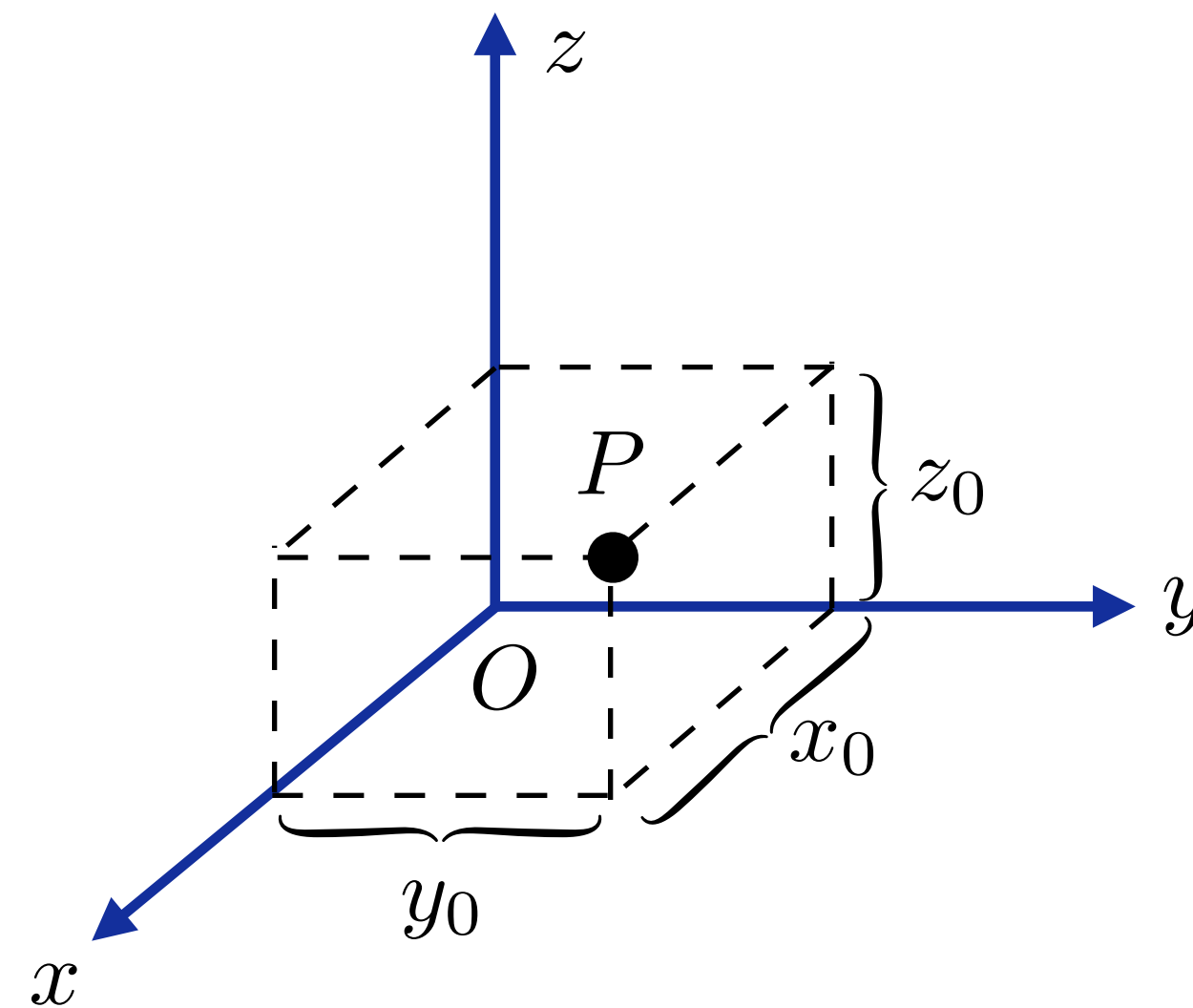
Mathematics of Computational Science
Department of Applied Mathematics
University of Twente

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Three-Dimensional Cartesian Coordinates

Three-dimensional space: Through any point in space, exactly three mutually perpendicular lines can pass, and three numbers are required to locate a point relative to an origin.



$P(x_0, y_0, z_0)$: point P in 3D space with coordinates x_0, y_0, z_0 along the x -, y -, and z -axes

Cartesian coordinate system:

- Defined by three mutually perpendicular axes through the origin: x -, y -, and z -axes.
- x - and y -axes lie in a “horizontal plane”; z -axis is “vertical”.
- Right-handed orientation: A *right-handed screw* turns from x to y to advance along z .

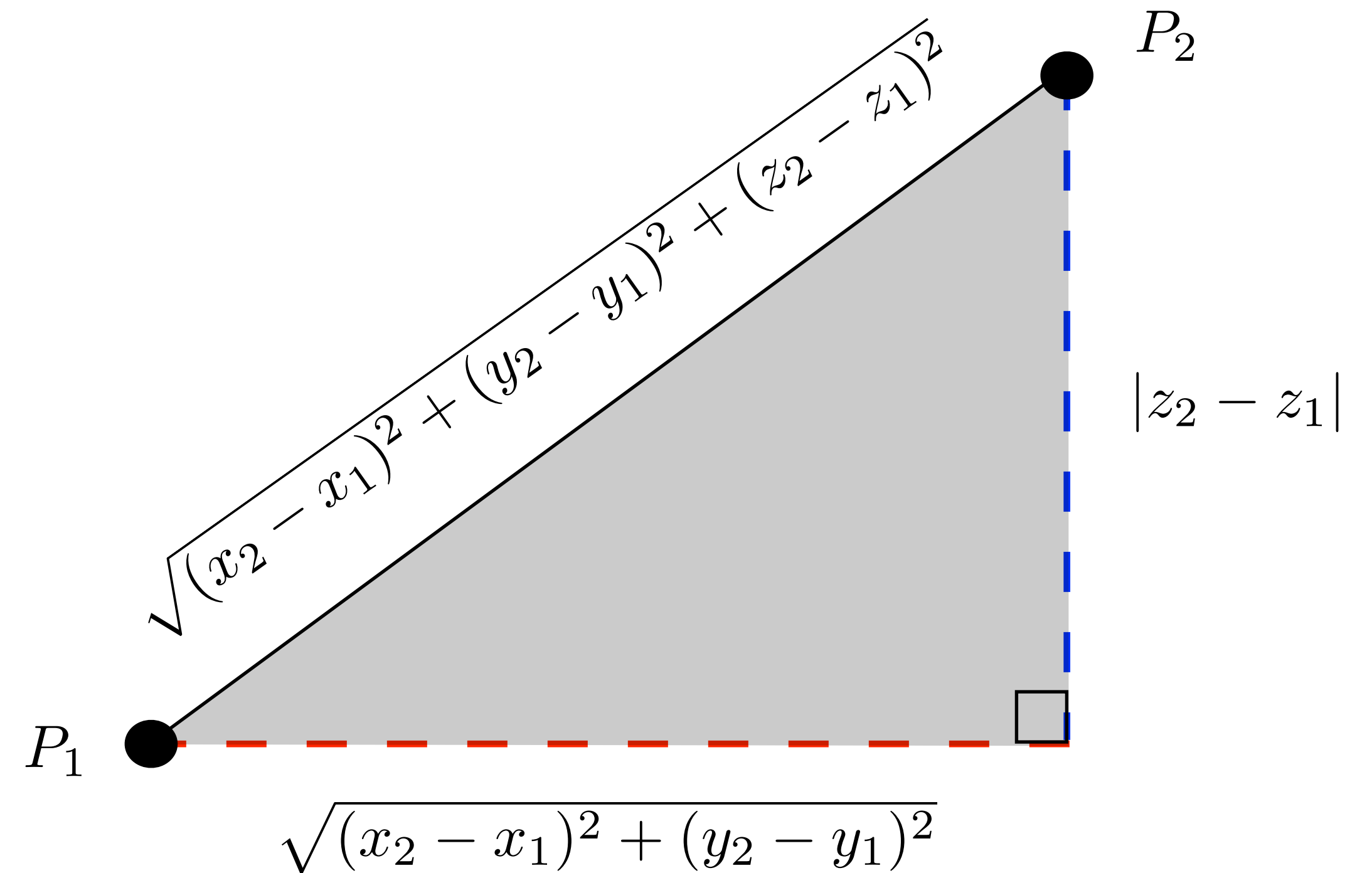
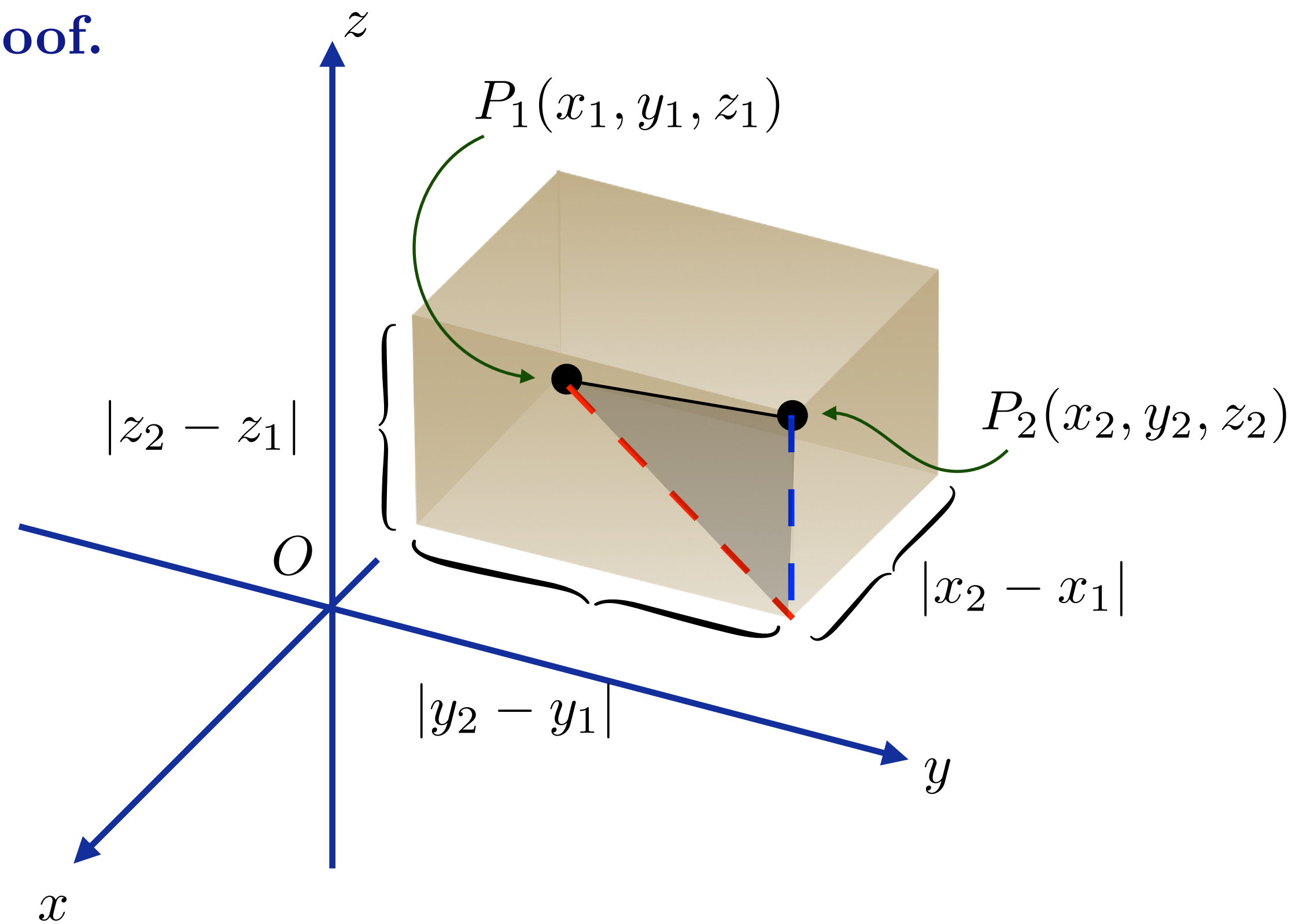
Coordinates of a point: Represented as (x_0, y_0, z_0) = signed distances along the x -, y -, and z -axes.

Distance and Spheres in Space

Theorem: The Distance between $P_1 (x_1, y_1, z_1)$ and $P_2 (x_2, y_2, z_2)$ is given by

$$|P_1 P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Proof.



Equation of a Sphere

A sphere is the set of all points in space that are the same distance $a > 0$ from a fixed point $P_0(x_0, y_0, z_0)$ (the center).

Equation: A point $P(x, y, z)$ lies on the sphere iff

$$|P_0P| = a,$$

which gives

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2.$$

Standard Equation:

- Center: (x_0, y_0, z_0)
- Radius: a

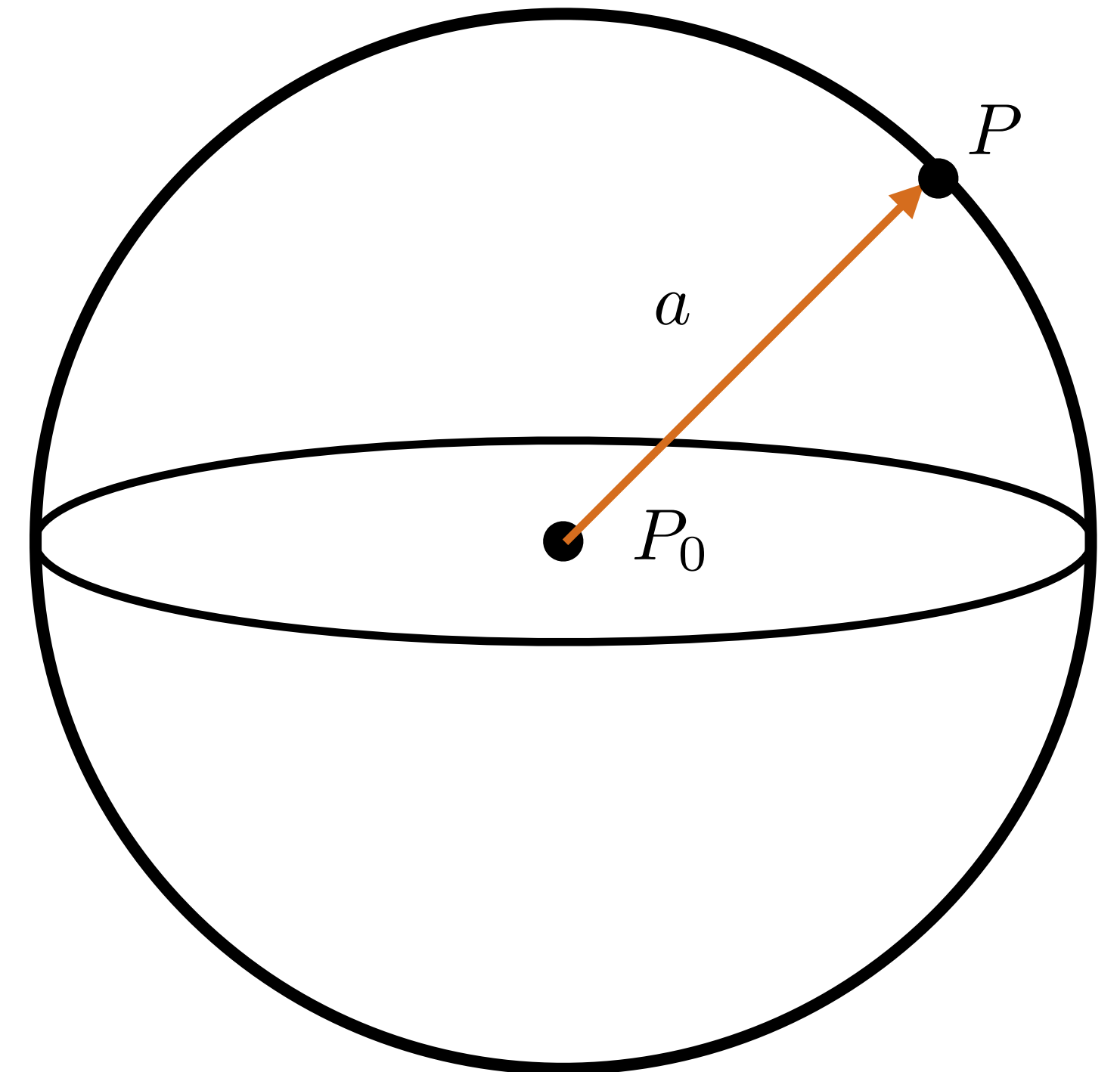
Example: Find the center and radius of the sphere

$$x^2 + y^2 + z^2 + 3x - 4z + 1 = 0$$

Solution. Complete the squares: $x^2 + 3x = (x + \frac{3}{2})^2 - \frac{9}{4}$ and $z^2 - 4z = (z - 2)^2 - 4$.

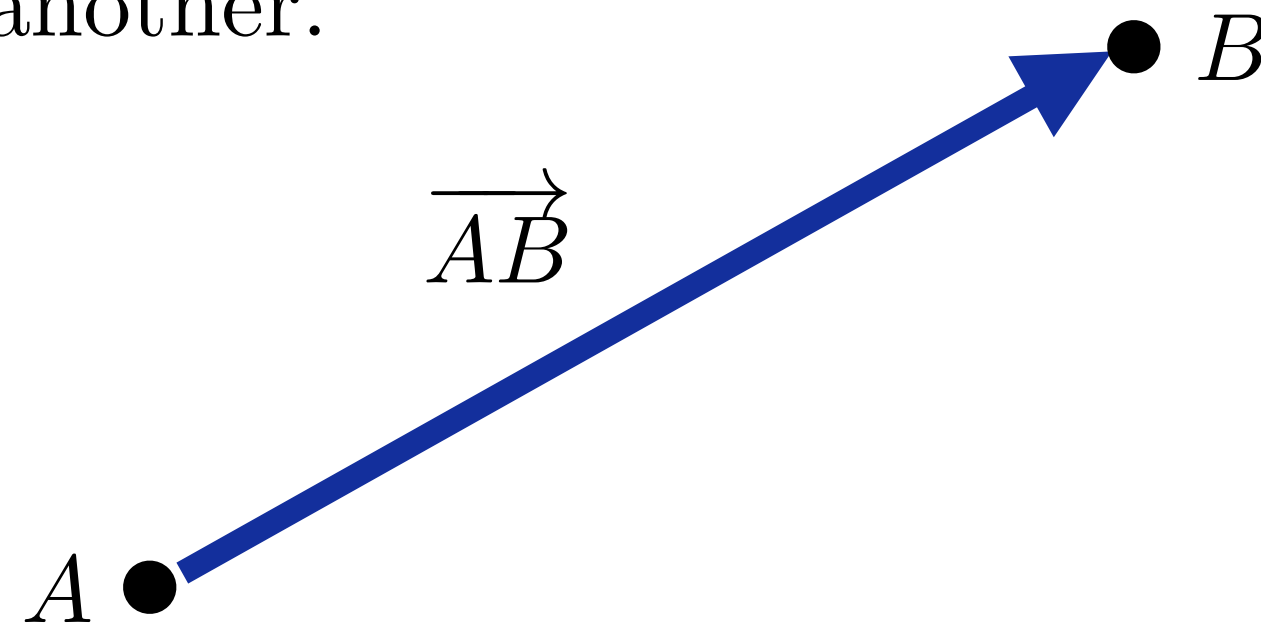
Thus, $(x + \frac{3}{2})^2 + y^2 + (z - 2)^2 = \frac{21}{4}$.

Center: $(-\frac{3}{2}, 0, 2)$, Radius: $\frac{\sqrt{21}}{2}$.



Vectors and Component Form

A vector is a quantity that has both **magnitude** (length) and **direction**, and is often represented as a directed line segment from one point to another.



Definition: The vector represented by the directed line segment $\overrightarrow{AB}$ has initial point A , terminal point B , and length $|\overrightarrow{AB}|$. Two vectors are equal if they have the same length and direction.

Definition: If $\mathbf{v}$ is a two-dimensional vector in the plane equal to the vector with initial point at the origin and terminal point (v_1, v_2) , then the component form of $\mathbf{v}$ is

$$\mathbf{v} = \langle v_1, v_2 \rangle.$$

If $\mathbf{v}$ is a three-dimensional vector equal to the vector with initial point at the origin and terminal point (v_1, v_2, v_3) , then the component form of $\mathbf{v}$ is

$$\mathbf{v} = \langle v_1, v_2, v_3 \rangle.$$

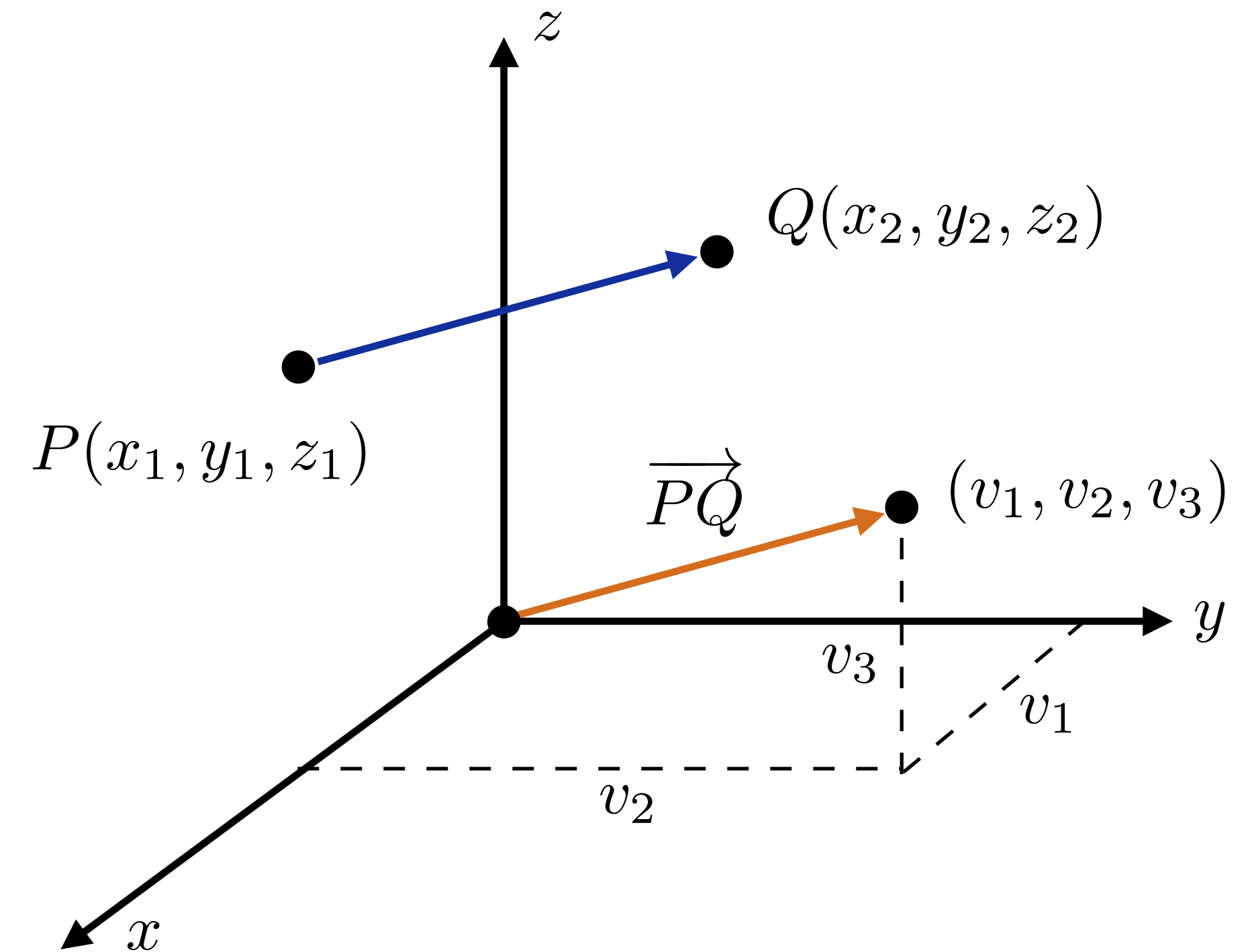
Magnitude of a Vector

Components of $\overrightarrow{PQ}$:

$$v_1 = x_2 - x_1, \quad v_2 = y_2 - y_1, \quad v_3 = z_2 - z_1.$$

Standard position vector:

$$\mathbf{v} = \overrightarrow{PQ} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$



Definition: The magnitude or length of the vector $\mathbf{v} = \overrightarrow{PQ}$ is the nonnegative number

$$|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Vectors Operations

Definition: Let $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ be vectors, and let $k \in \mathbb{R}$ be a scalar.

Addition: $\mathbf{u} + \mathbf{v} = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3 \rangle,$

Scalar multiplication: $k\mathbf{u} = \langle ku_1, ku_2, ku_3 \rangle.$

Example: Let $\mathbf{u} = \langle -1, 3, 1 \rangle$ and $\mathbf{v} = \langle 4, 7, 0 \rangle$. Find the components of

$$(a) \ 2\mathbf{u} + 3\mathbf{v}, \quad (b) \ \mathbf{u} - \mathbf{v}, \quad (c) \ \left| \frac{1}{2}\mathbf{u} \right|$$

Properties of Vector Operations: Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors and a, b be scalars. The following properties hold:

- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (Commutative)
- $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (Associative)
- $\mathbf{u} + \mathbf{0} = \mathbf{u}$ (Additive identity)
- $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$ (Additive inverse)
- $0\mathbf{u} = \mathbf{0}$ (Zero scalar)
- $1\mathbf{u} = \mathbf{u}$ (Multiplicative identity)
- $a(b\mathbf{u}) = (ab)\mathbf{u}$ (Associativity)
- $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$ (Distributive)
- $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$ (Distributive)

Dot Product and Angle Between Vectors

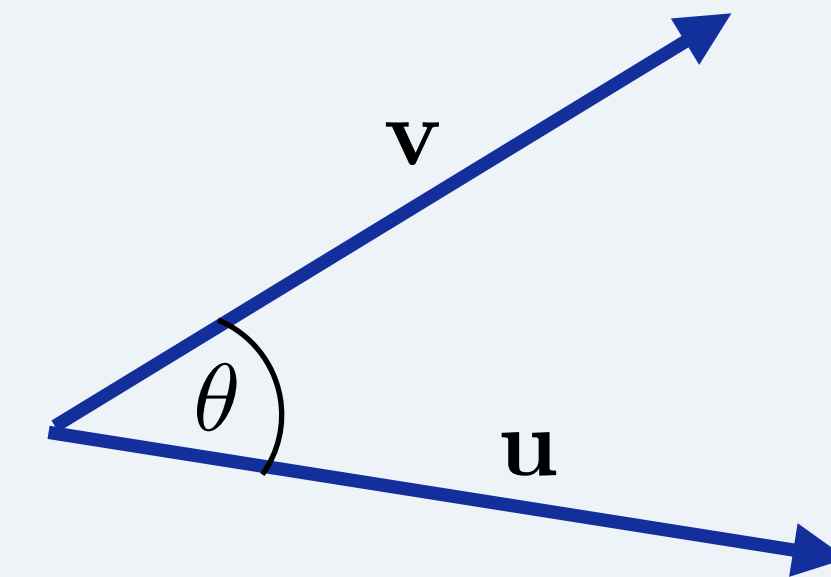
Definition: The dot product $\mathbf{u} \cdot \mathbf{v}$ (“u dot v”) of vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ is

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3$$

Theorem: The angle θ between $\mathbf{u}$ and $\mathbf{v}$ is defined by

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}, \quad 0 \leq \theta \leq \pi$$

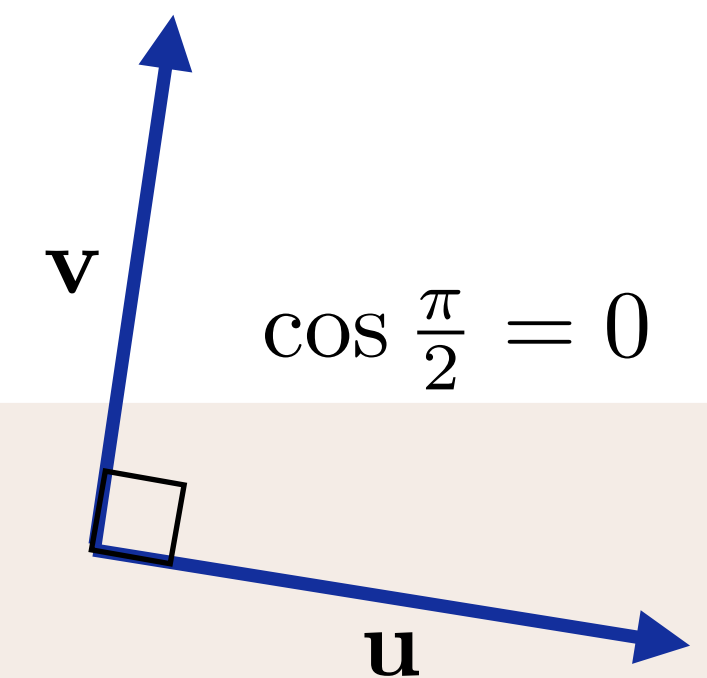
where $\mathbf{u} \cdot \mathbf{v}$ is the dot product and $|\mathbf{u}|, |\mathbf{v}|$ are the magnitudes of the vectors.



Special cases:

- $\theta = 0$ if $\mathbf{u}$ and $\mathbf{v}$ point in the same direction.
- $\theta = \pi$ if $\mathbf{u}$ and $\mathbf{v}$ point in opposite directions.

Definition: Vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal (or perpendicular) if and only if $\mathbf{u} \cdot \mathbf{v} = 0$.



Properties of the Dot Product

Properties of the Dot Product

Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors and c a scalar. Then:

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ (Commutative)
2. $(c\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (c\mathbf{v}) = c(\mathbf{u} \cdot \mathbf{v})$ (Scalar multiplication)
3. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$ (Distributive)
4. $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$ (Magnitude squared)
5. $\mathbf{0} \cdot \mathbf{u} = 0$ (Zero vector)

Proof (4): Let $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$. By the definition of the dot product, $\mathbf{u} \cdot \mathbf{u} = u_1^2 + u_2^2 + u_3^2$. The magnitude of $\mathbf{u}$ is $|\mathbf{u}| = \sqrt{u_1^2 + u_2^2 + u_3^2}$. Squaring both sides gives $|\mathbf{u}|^2 = u_1^2 + u_2^2 + u_3^2$. Comparing with the dot product, we conclude $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$.

Vector Projection

Vector projection: The projection of a vector $\mathbf{u} = \overrightarrow{PQ}$ onto a nonzero vector $\mathbf{v} = \overrightarrow{PS}$ is the unique vector $\overrightarrow{PR}$ parallel to $\mathbf{v}$ such that the vector $\overrightarrow{QR}$ is perpendicular to $\mathbf{v}$. Formally, this projection is denoted

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \overrightarrow{PR}.$$

Formula: Let θ be the angle between $\mathbf{u}$ and $\mathbf{v}$.

$$\text{proj}_{\mathbf{v}} \mathbf{u} = (|\mathbf{u}| \cos \theta) \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \mathbf{v}.$$

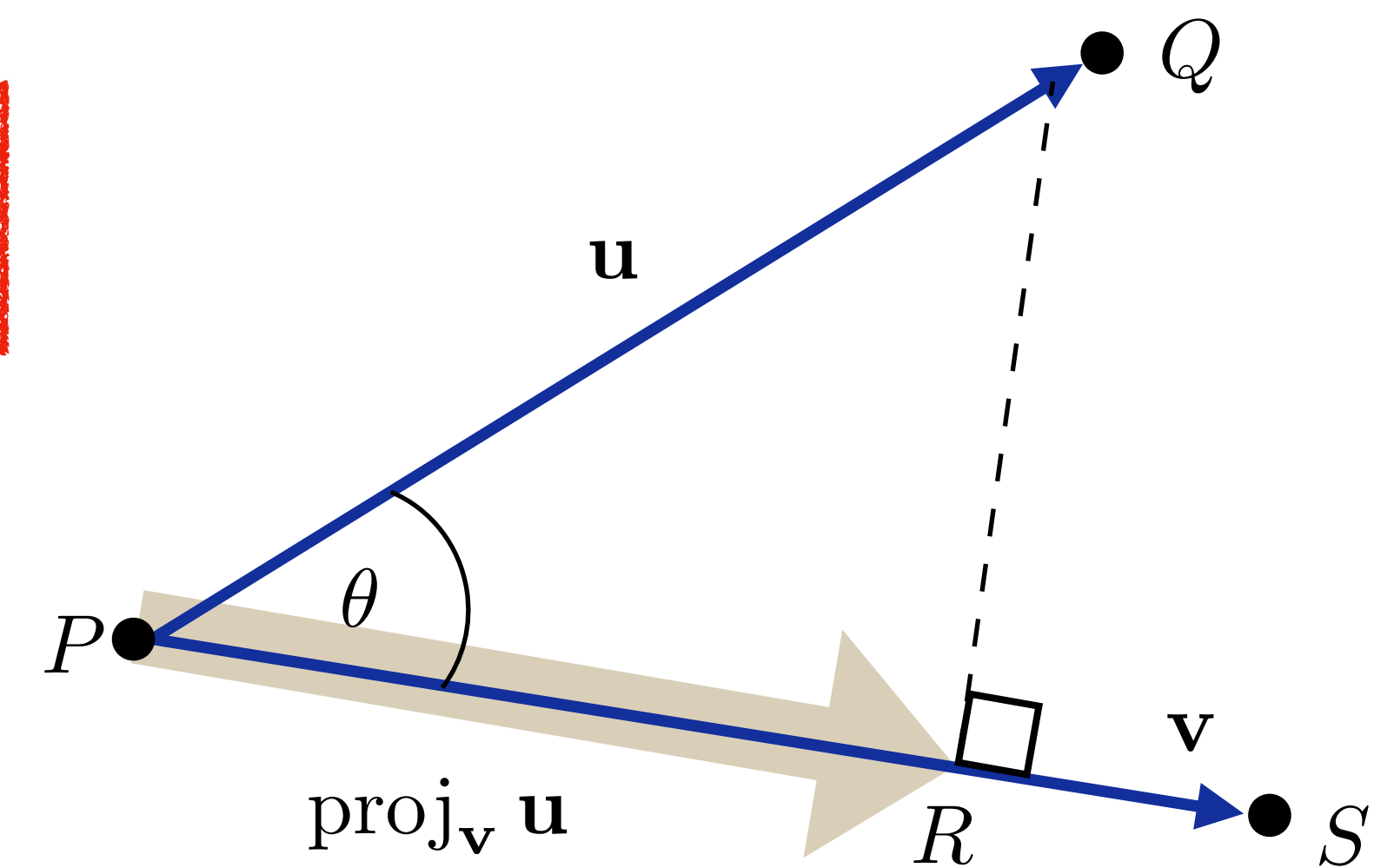
Notes:

- If θ is acute, projection points along $\mathbf{v}$.
- If θ is obtuse, projection points opposite to $\mathbf{v}$.

The scalar component of $\mathbf{u}$ in the direction of $\mathbf{v}$ is the scalar

$$|\mathbf{u}| \cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|} = \mathbf{u} \cdot \frac{\mathbf{v}}{|\mathbf{v}|}$$

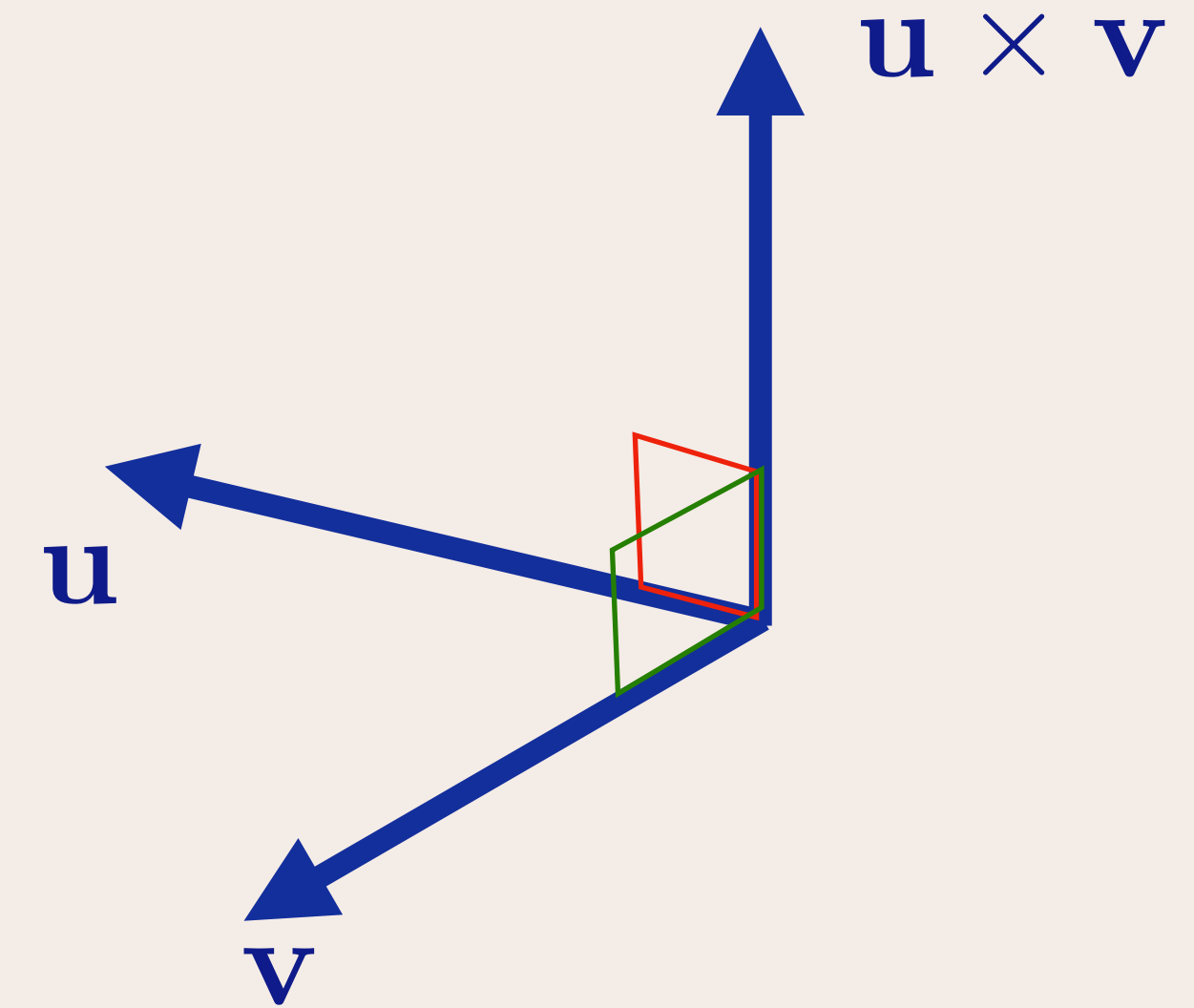
Example: Find the vector projection of $\mathbf{u} = 6\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$ onto $\mathbf{v} = \mathbf{i} - 2\mathbf{j} - 2\mathbf{k}$ and the scalar component of $\mathbf{u}$ in the direction of $\mathbf{v}$.



The Cross Product

Definition (Cross Product): For any vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^3$, the cross product $\mathbf{u} \times \mathbf{v}$ is the unique vector satisfying:

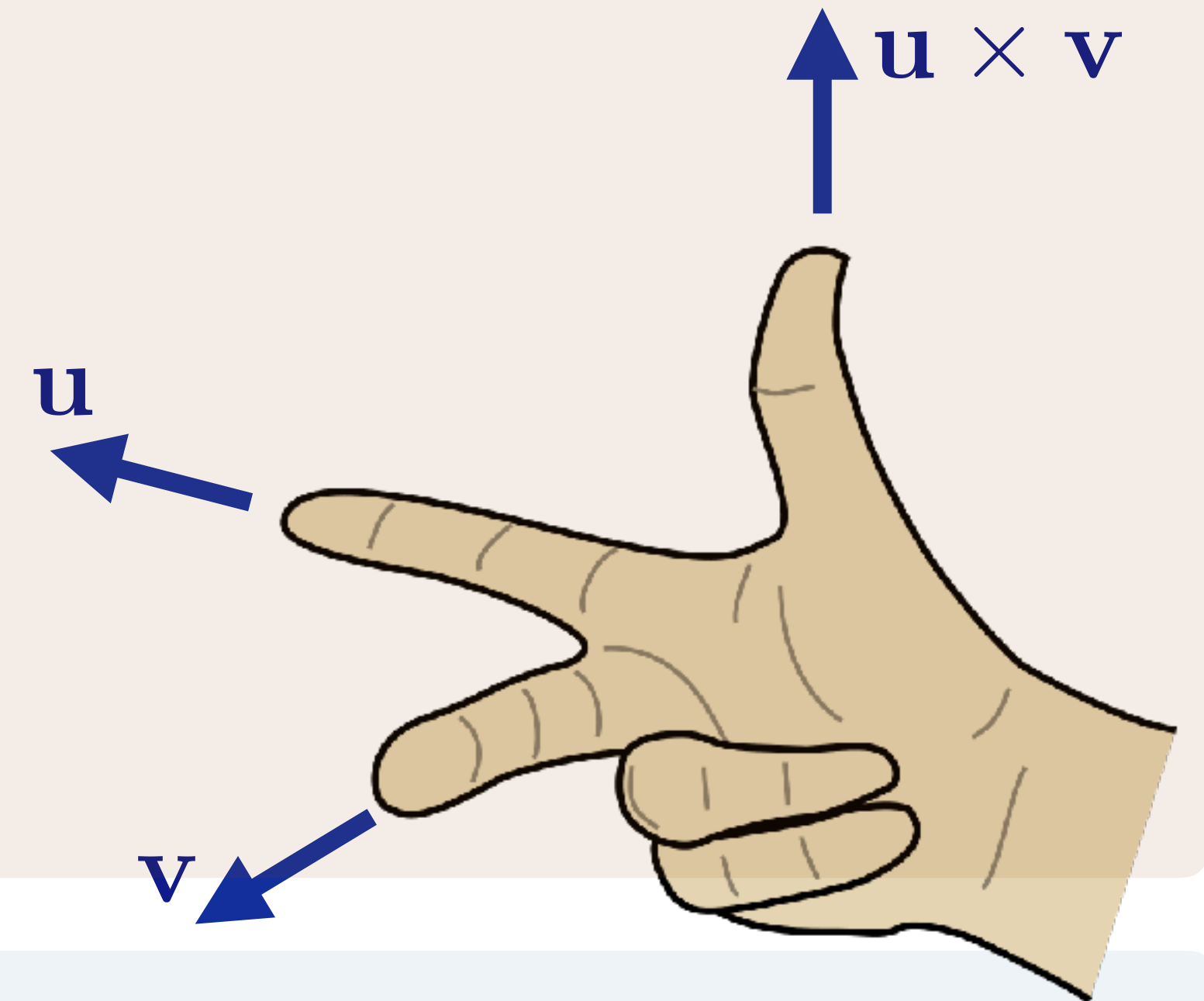
1. $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = 0$ and $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} = 0$,
2. $|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}| |\mathbf{v}| \sin \theta$, where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$,
3. $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{u} \times \mathbf{v}$ form a right-handed triad.



The Cross Product

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3. $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{u} \times \mathbf{v}$ form a right-handed triad.



Components of the Cross Product: If

$$\mathbf{u} = u_1 \mathbf{i} + u_2 \mathbf{j} + u_3 \mathbf{k}, \quad \mathbf{v} = v_1 \mathbf{i} + v_2 \mathbf{j} + v_3 \mathbf{k},$$

then the cross product is given by

$$\mathbf{u} \times \mathbf{v} = (u_2 v_3 - u_3 v_2) \mathbf{i} + (u_3 v_1 - u_1 v_3) \mathbf{j} + (u_1 v_2 - u_2 v_1) \mathbf{k}.$$

The Cross Product

Example

1. Verify that:

$$\begin{array}{lll} \mathbf{i} \times \mathbf{i} = \mathbf{0}, & \mathbf{i} \times \mathbf{j} = \mathbf{k}, & \mathbf{j} \times \mathbf{i} = -\mathbf{k}, \\ \mathbf{j} \times \mathbf{j} = \mathbf{0}, & \mathbf{j} \times \mathbf{k} = \mathbf{i}, & \mathbf{k} \times \mathbf{j} = -\mathbf{i}, \\ \mathbf{k} \times \mathbf{k} = \mathbf{0}, & \mathbf{k} \times \mathbf{i} = \mathbf{j}, & \mathbf{i} \times \mathbf{k} = -\mathbf{j}. \end{array}$$

2. Evaluate $(2\mathbf{i} + \mathbf{j} - 3\mathbf{k}) \times (-2\mathbf{j} + 5\mathbf{k})$.

Solution.

$$\begin{aligned} (2\mathbf{i} + \mathbf{j} - 3\mathbf{k}) \times (-2\mathbf{j} + 5\mathbf{k}) &= ((1)(5) - (-2)(-3))\mathbf{i} + ((-3)(0) - (2)(5))\mathbf{j} \\ &\quad + ((2)(-2) - (1)(0))\mathbf{k} \\ &= -\mathbf{i} - 10\mathbf{j} - 4\mathbf{k}. \end{aligned}$$

The Cross Product

Properties of the Cross Product: Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors and r, s be scalars. Then:

1. $(r\mathbf{u}) \times (s\mathbf{v}) = (rs)(\mathbf{u} \times \mathbf{v})$ (Scalar multiplication)
2. $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$ (Distributive over addition)
3. $\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v})$ (Anticommutative)
4. $(\mathbf{v} + \mathbf{w}) \times \mathbf{u} = \mathbf{v} \times \mathbf{u} + \mathbf{w} \times \mathbf{u}$ (Distributive over addition)
5. $\mathbf{0} \times \mathbf{u} = \mathbf{0}$ (Zero vector)
6. $\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$ (Triple product)

Theorem (Parallel Vectors): Nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are parallel if and only if $\mathbf{u} \times \mathbf{v} = \mathbf{0}$.

Proof. ($\Rightarrow$) Suppose $\mathbf{u}$ and $\mathbf{v}$ are parallel. Then $\mathbf{v} = k\mathbf{u}$ for some scalar k . Hence,

$$\mathbf{u} \times \mathbf{v} = \mathbf{u} \times (k\mathbf{u}) = k(\mathbf{u} \times \mathbf{u}) = k\mathbf{0} = \mathbf{0}.$$

($\Leftarrow$) Conversely, suppose $\mathbf{u} \times \mathbf{v} = \mathbf{0}$. By the definition of the cross product, $|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}| |\mathbf{v}| \sin \theta$, where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$. Since $\mathbf{u}, \mathbf{v} \neq \mathbf{0}$, the equality $\mathbf{u} \times \mathbf{v} = \mathbf{0}$ implies $\sin \theta = 0$, so $\theta = 0$ or $\theta = \pi$. Therefore, $\mathbf{u}$ and $\mathbf{v}$ are parallel.

Area of a Parallelogram

Theorem: If $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$, then the area of the parallelogram determined by $\mathbf{u}$ and $\mathbf{v}$ is

$$A = |\mathbf{u} \times \mathbf{v}|.$$

Proof.

- The base of the parallelogram has length $|\mathbf{u}|$.
- The height is $|\mathbf{v}| \sin \theta$, where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$.
- Thus,

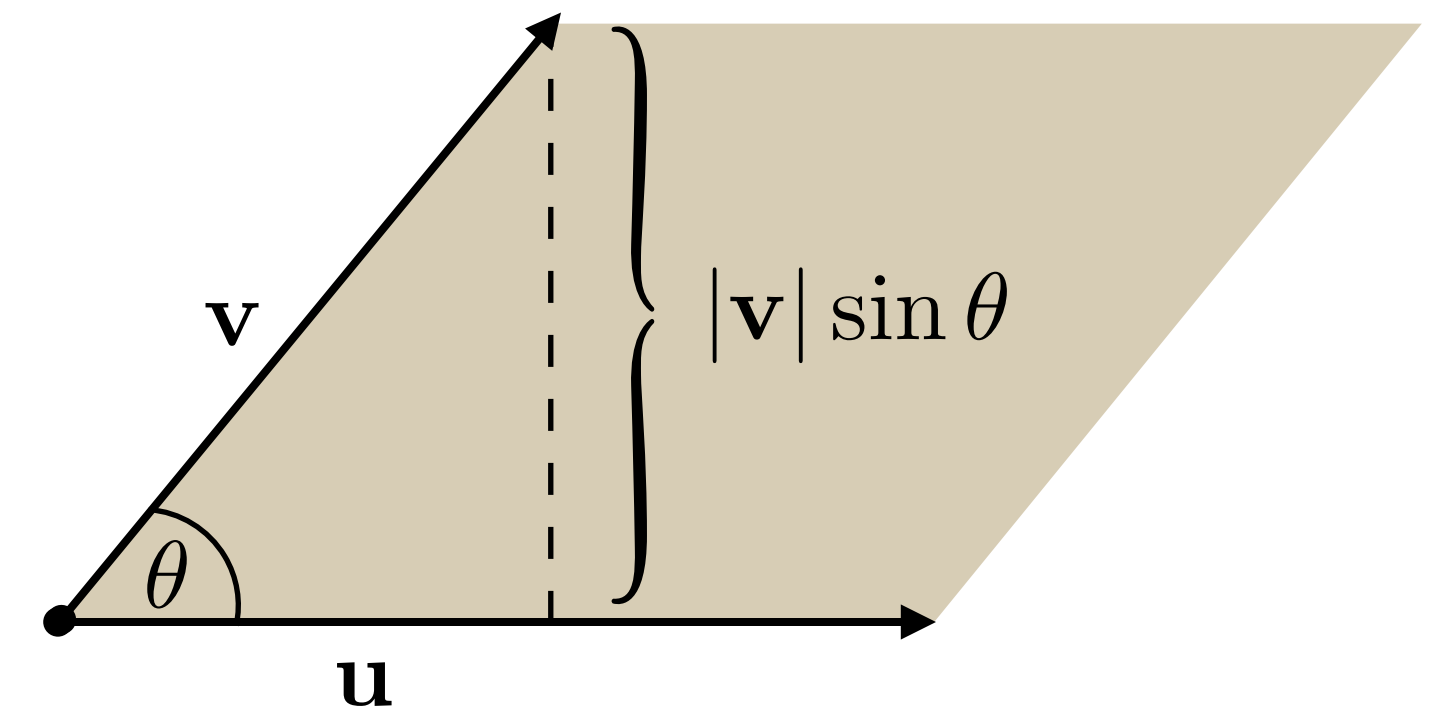
$$A = |\mathbf{u}| |\mathbf{v}| \sin \theta.$$

- By definition of the cross product,

$$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}| |\mathbf{v}| \sin \theta.$$

Conclusion:

$$A = |\mathbf{u} \times \mathbf{v}|$$



Determinants and Cross Product

Determinant of a 2×2 matrix:

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

Determinant of a 3×3 matrix:

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a \det \begin{bmatrix} e & f \\ h & i \end{bmatrix} - b \det \begin{bmatrix} d & f \\ g & i \end{bmatrix} + c \det \begin{bmatrix} d & e \\ g & h \end{bmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Cross product as a determinant: For $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ and $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$,

$$\mathbf{u} \times \mathbf{v} = \det \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{bmatrix}$$

Expanding along the first row:

$$\begin{aligned} \mathbf{u} \times \mathbf{v} &= \mathbf{i} \det \begin{bmatrix} u_2 & u_3 \\ v_2 & v_3 \end{bmatrix} - \mathbf{j} \det \begin{bmatrix} u_1 & u_3 \\ v_1 & v_3 \end{bmatrix} + \mathbf{k} \det \begin{bmatrix} u_1 & u_2 \\ v_1 & v_2 \end{bmatrix} \\ &= (u_2v_3 - u_3v_2)\mathbf{i} - (u_1v_3 - u_3v_1)\mathbf{j} + (u_1v_2 - u_2v_1)\mathbf{k} \\ &= (u_2v_3 - u_3v_2)\mathbf{i} + (u_3v_1 - u_1v_3)\mathbf{j} + (u_1v_2 - u_2v_1)\mathbf{k}. \end{aligned}$$

same as in previous slide

Determinants and Cross Product

Example: Find $\mathbf{u} \times \mathbf{v}$ and $\mathbf{v} \times \mathbf{u}$ if $\mathbf{u} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{v} = -4\mathbf{i} + 3\mathbf{j} + \mathbf{k}$.

Solution.

$$\mathbf{u} \times \mathbf{v} = \det \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 1 \\ -4 & 3 & 1 \end{bmatrix} = -2\mathbf{i} - 6\mathbf{j} + 10\mathbf{k}.$$

Since $\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v})$, we get

$$\mathbf{v} \times \mathbf{u} = 2\mathbf{i} + 6\mathbf{j} - 10\mathbf{k}.$$

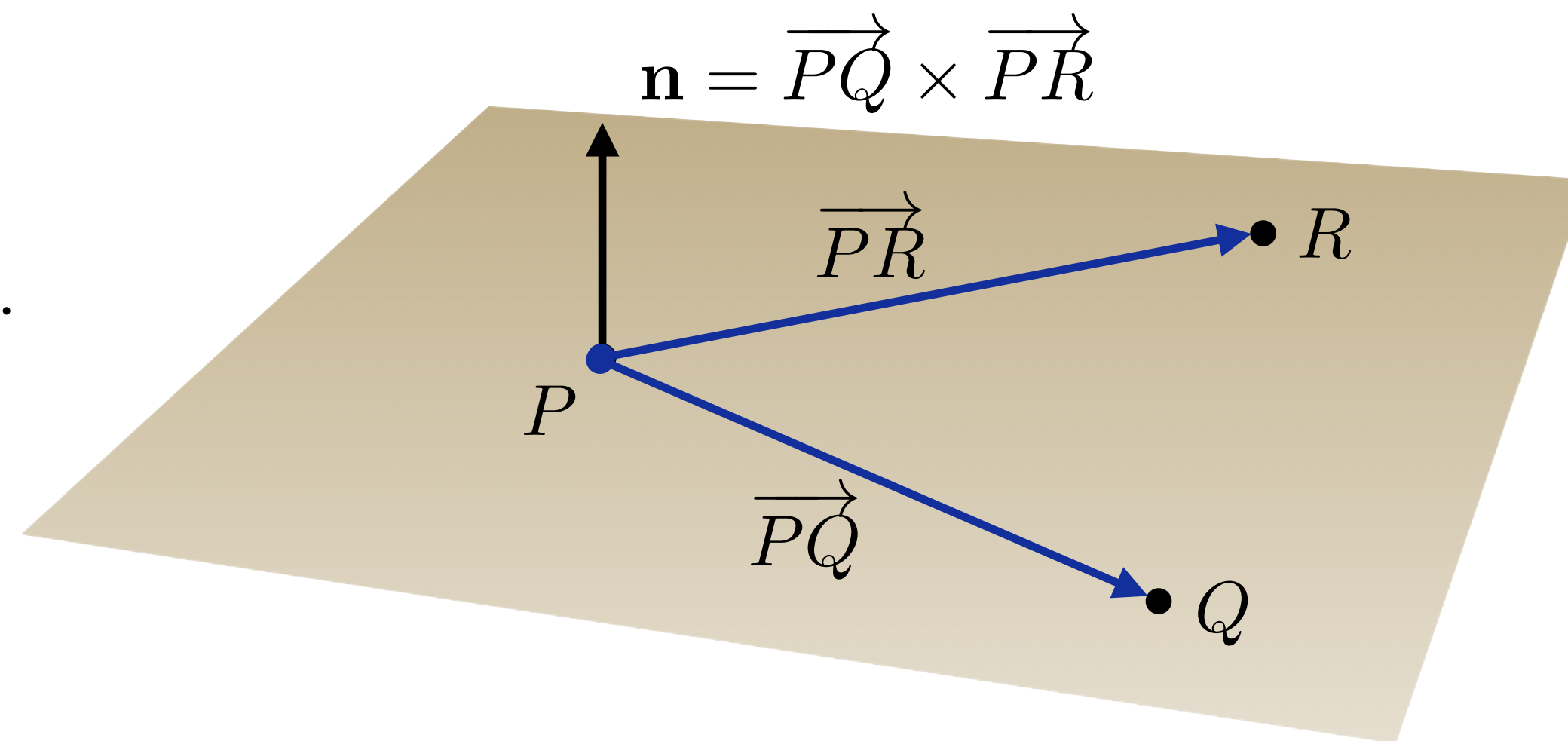
Example: Find a vector perpendicular to the plane of $P(1, -1, 0)$, $Q(2, 1, -1)$, and $R(-1, 1, 2)$.

Solution.

$$\overrightarrow{PQ} = (2 - 1)\mathbf{i} + (1 - (-1))\mathbf{j} + (-1 - 0)\mathbf{k} = \mathbf{i} + 2\mathbf{j} - \mathbf{k},$$

$$\overrightarrow{PR} = (-1 - 1)\mathbf{i} + (1 - (-1))\mathbf{j} + (2 - 0)\mathbf{k} = -2\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}.$$

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \det \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -1 \\ -2 & 2 & 2 \end{bmatrix} = 6\mathbf{i} - 0\mathbf{j} + 6\mathbf{k}.$$



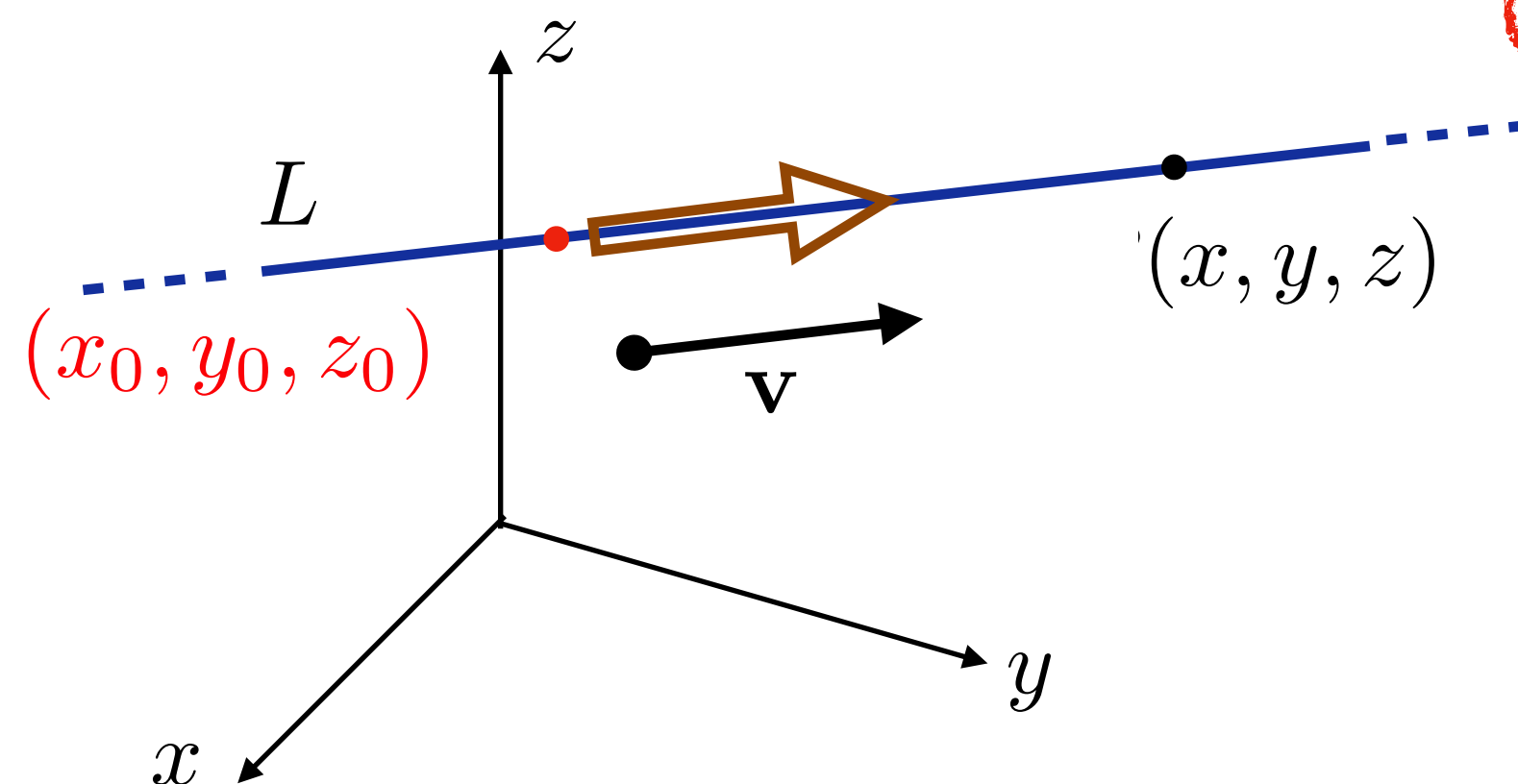
Therefore, the vector $\mathbf{n} = 6\mathbf{i} + 6\mathbf{k}$ is perpendicular to the plane.

Lines and Planes in Space

In the plane ($\mathbb{R}^2$): A line is determined by a point and a slope

In space ($\mathbb{R}^3$): A line is determined by a point and a direction vector.

point: $P_0(x_0, y_0)$;
slope: m ;
line equation: $y = (x - x_0)m + y_0$



Vector Equation for a Line: A vector equation for the line L through (x_0, y_0, z_0) parallel to $\mathbf{v}$ is

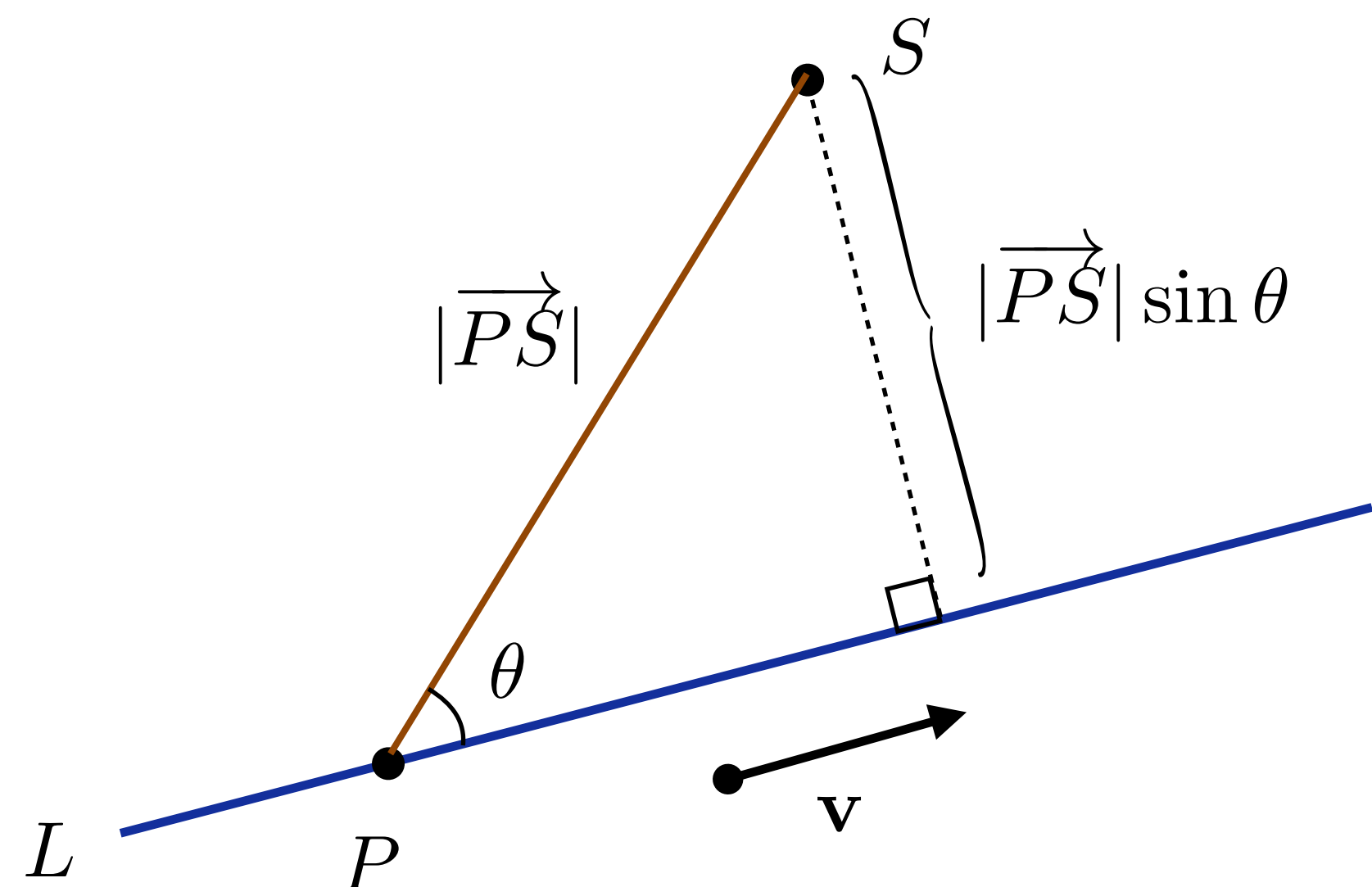
$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}, \quad -\infty < t < \infty$$

where $\mathbf{r}$ is the position vector of a point (x, y, z) on L and $\mathbf{r}_0$ is the position vector of (x_0, y_0, z_0) .

Parametric Equations for a Line: The standard parametrization of the line through (x_0, y_0, z_0) parallel to $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$ is

$$x = x_0 + tv_1, \quad y = y_0 + tv_2, \quad z = z_0 + tv_3, \quad -\infty < t < \infty$$

The Distance from a Point to a Line in Space



Distance from a point S

to a line through P parallel to $\mathbf{v}$: $d = \frac{|\vec{PS} \times \mathbf{v}|}{|\mathbf{v}|}$

Example: Find the distance from the point $S(1, 1, 5)$ to the line: $L: x = 1 + t, y = 3 - t, z = 2t$.

Solution. The line is given by: $\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}$, $\mathbf{r}_0 = 1\mathbf{i} + 3\mathbf{j} + 0\mathbf{k}$, $\mathbf{v} = 1\mathbf{i} - 1\mathbf{j} + 2\mathbf{k}$.

Vector from the point $\mathbf{r}_0$ on the line, given by $P(1, 3, 0)$, on the line to S is

$$\vec{PS} = (1 - 1)\mathbf{i} + (1 - 3)\mathbf{j} + (5 - 0)\mathbf{k} = -2\mathbf{j} + 5\mathbf{k}.$$

$$\text{Compute the cross product: } \vec{PS} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -2 & 5 \\ 1 & -1 & 2 \end{vmatrix} = 1\mathbf{i} + 5\mathbf{j} + 2\mathbf{k} \implies |\vec{PS} \times \mathbf{v}| = \sqrt{1^2 + 5^2 + 2^2} = \sqrt{30}.$$

$$\text{Since } |\mathbf{v}| = \sqrt{1^2 + (-1)^2 + 2^2} = \sqrt{6}, \text{ we conclude } d = \frac{|\vec{PS} \times \mathbf{v}|}{|\mathbf{v}|} = \frac{\sqrt{30}}{\sqrt{6}} = \sqrt{5}.$$

Equation for a Plane in Space

Plane in space: Determined by a point on the plane and a normal vector (perpendicular to the plane)

Suppose plane M passes through

$$P_0(x_0, y_0, z_0)$$

and is normal to

$$\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}.$$

Then any point $P(x, y, z)$ on the plane satisfies

$$\overrightarrow{P_0P} \cdot \mathbf{n} = 0$$

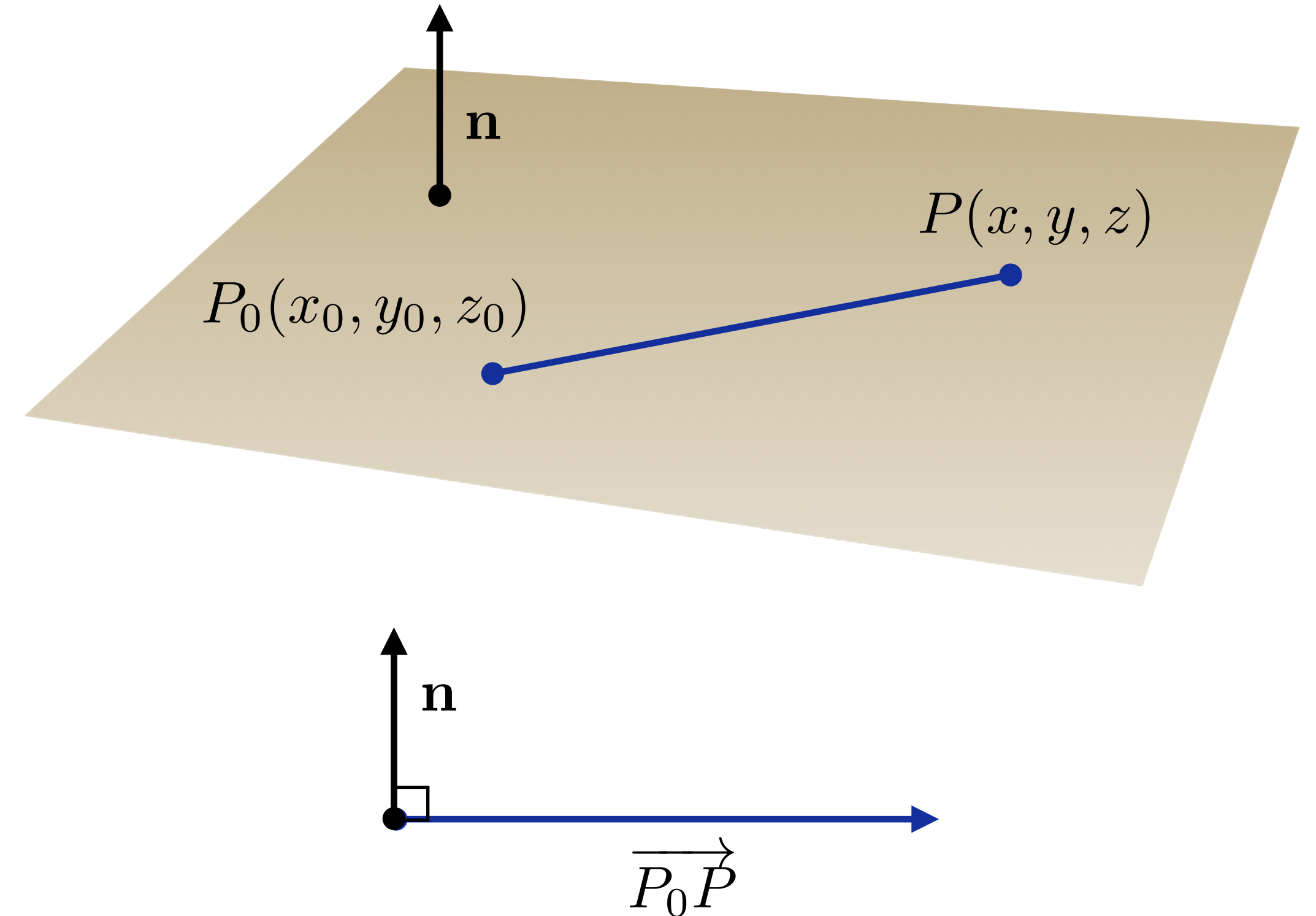
This equation is equivalent to

$$(A\mathbf{i} + B\mathbf{j} + C\mathbf{k}) \cdot [(x - x_0)\mathbf{i} + (y - y_0)\mathbf{j} + (z - z_0)\mathbf{k}] = 0$$

or

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

This can be simplified to: $Ax + By + Cz = D$, where $D = Ax_0 + By_0 + Cz_0$



Equation for a Plane in Space

Example: Find an equation for the plane through $P_0(-3, 0, 7)$ perpendicular to $\mathbf{n} = 5\mathbf{i} + 2\mathbf{j} - \mathbf{k}$.

Solution: A plane through $P_0(x_0, y_0, z_0)$ with normal vector $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ has equation

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0.$$

For $P_0(-3, 0, 7)$ and $\mathbf{n} = 5\mathbf{i} + 2\mathbf{j} - \mathbf{k}$:

$$5(x + 3) + 2y - (z - 7) = 0 \quad \implies \quad 5x + 2y - z = -22$$

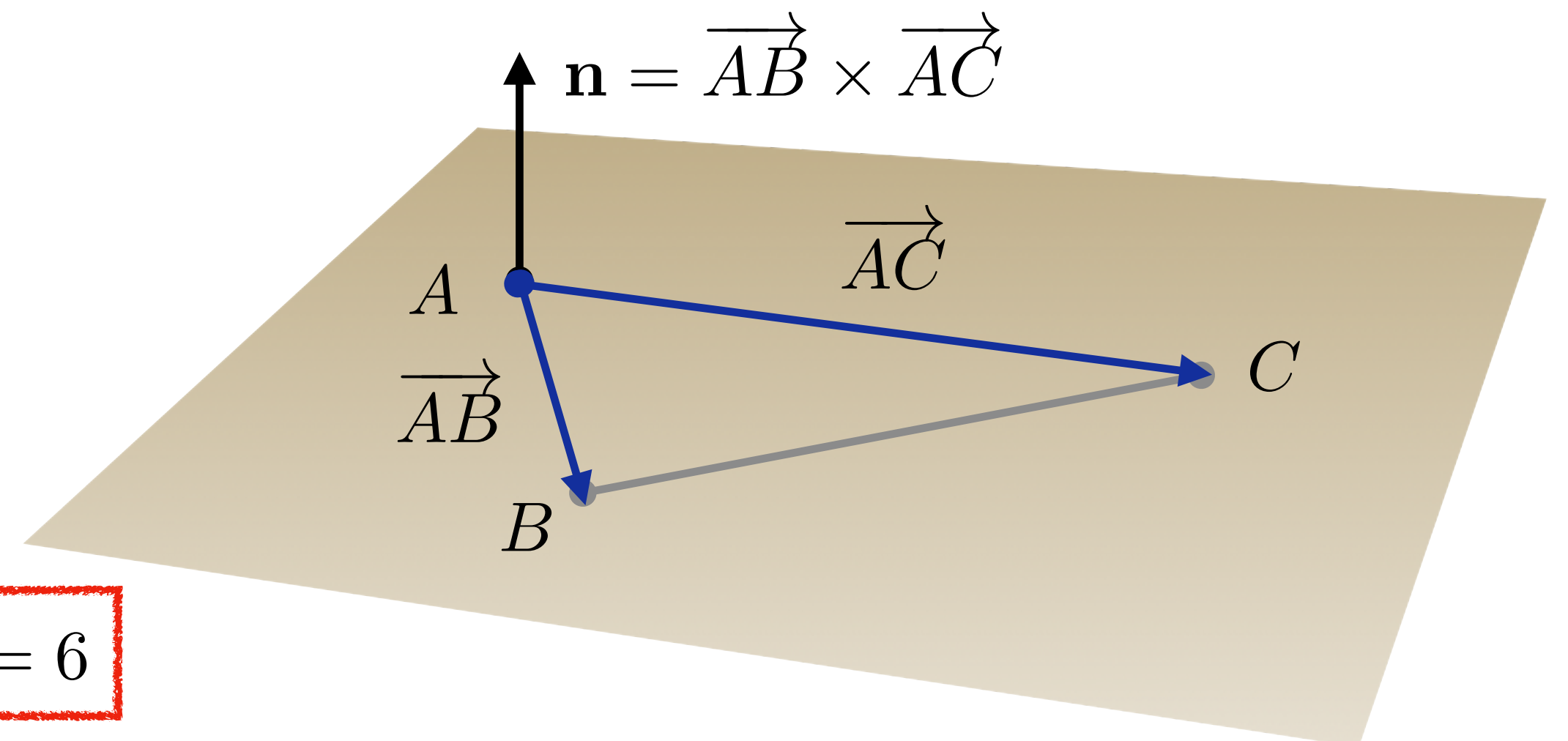
Example: Find an equation for the plane through $A(0, 0, 1)$, $B(2, 0, 0)$, and $C(0, 3, 0)$.

We find a vector normal to the plane:

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 0 & -1 \\ 0 & 3 & -1 \end{vmatrix} = 3\mathbf{i} + 2\mathbf{j} + 6\mathbf{k}$$

Using $A(0, 0, 1)$ as the point on the plane we get:

$$3(x - 0) + 2(y - 0) + 6(z - 1) = 0 \quad \implies \quad 3x + 2y + 6z = 6$$



Lines of Intersection

- Two planes are parallel if and only if their normal vectors are parallel, i.e.,

$$\mathbf{n}_1 = k\mathbf{n}_2$$

for some scalar k .

- Two planes that are not parallel intersect in a line.

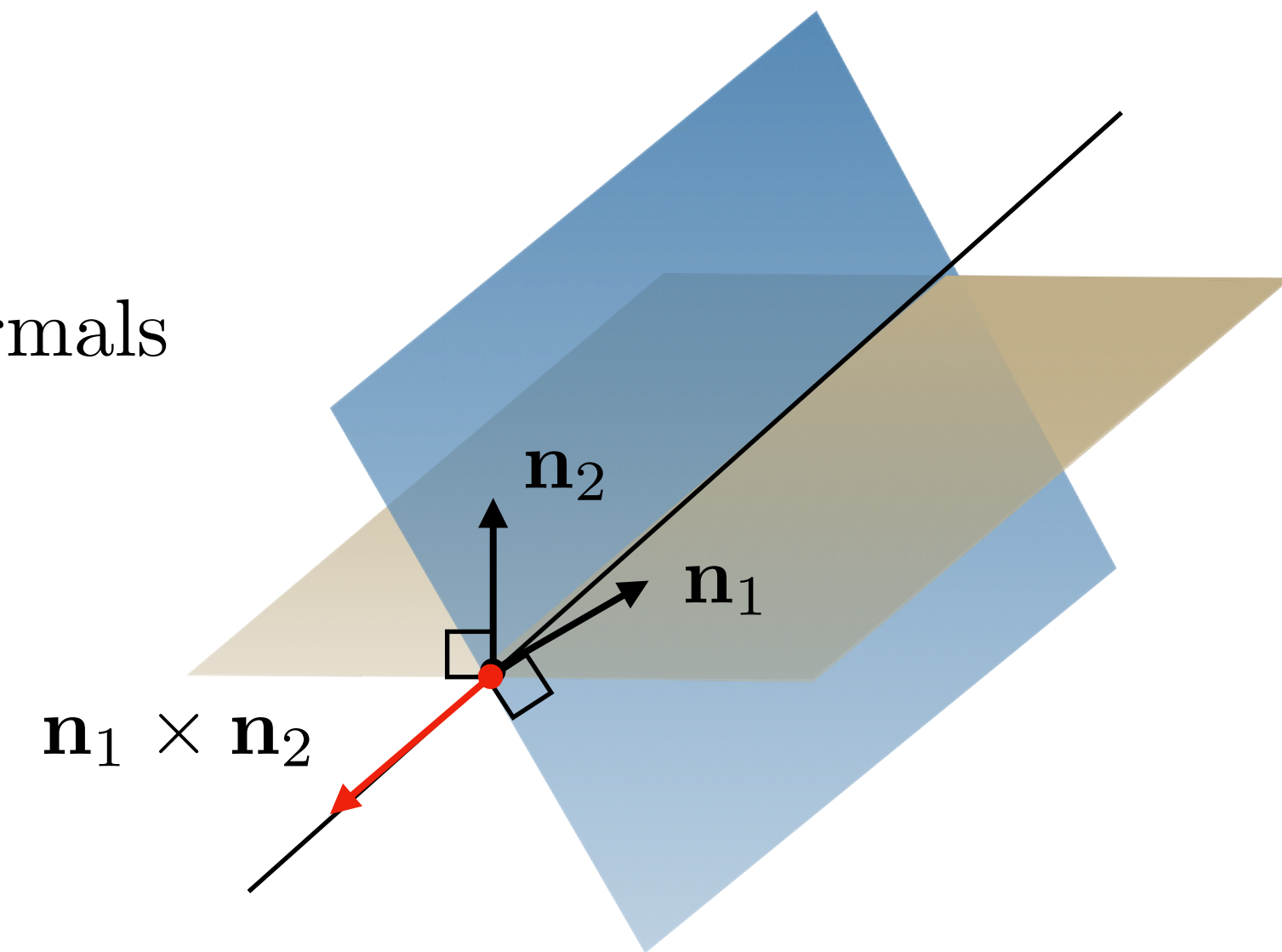
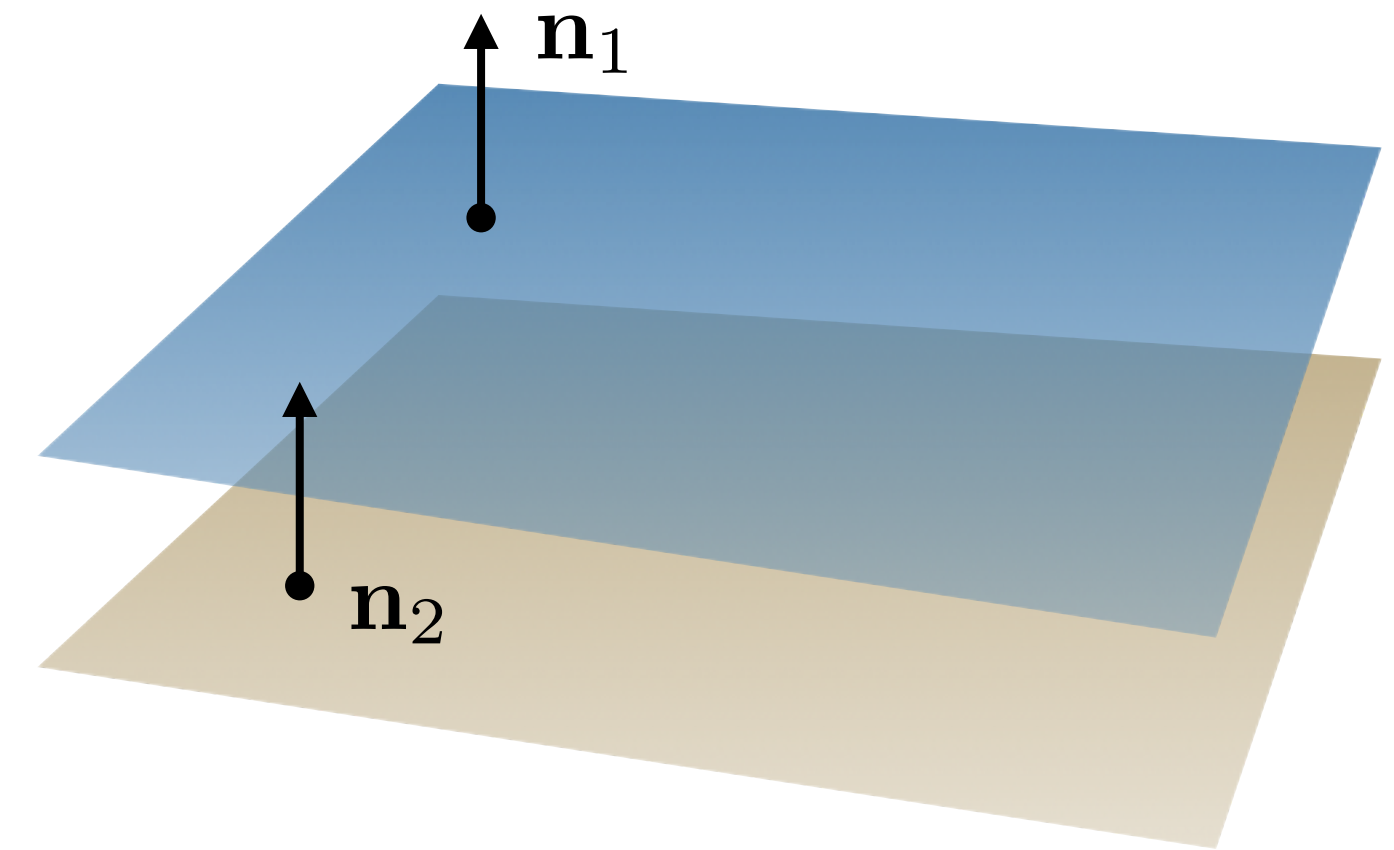
Example: Find a vector parallel to the line of intersection of the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

Solution: The line of intersection of two planes is perpendicular to both normals $\mathbf{n}_1$ and $\mathbf{n}_2$, so it is parallel to

$$\mathbf{n}_1 \times \mathbf{n}_2.$$

In this case,

$$\mathbf{n}_1 \times \mathbf{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -6 & -2 \\ 2 & 1 & -2 \end{vmatrix} = 14\mathbf{i} + 2\mathbf{j} + 15\mathbf{k}.$$

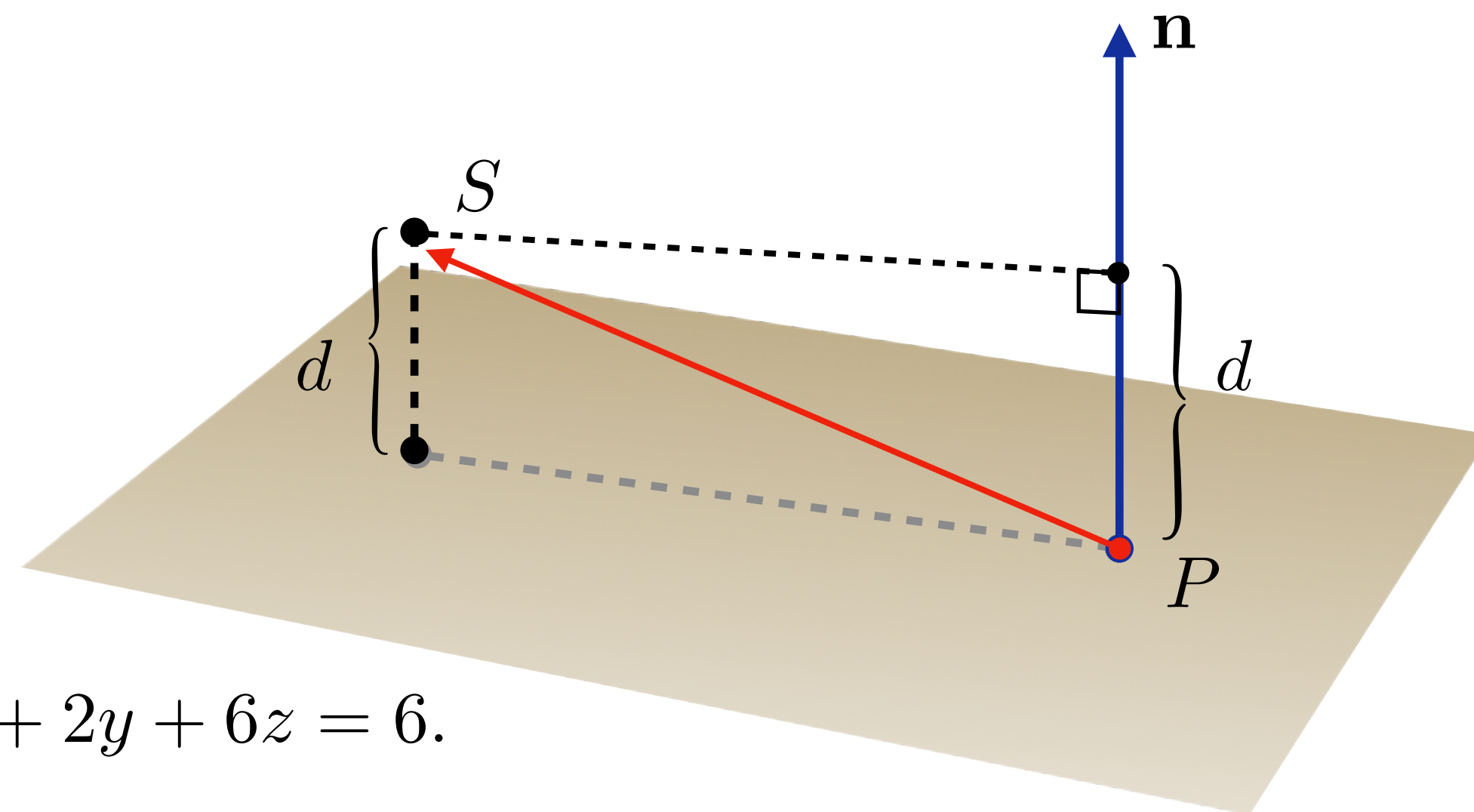


The Distance from a Point to a Plane

If P is a point on a plane with normal vector $\mathbf{n}$, the distance from any point S to the plane is the length of the projection of $\overrightarrow{PS}$ onto $\mathbf{n}$:

$$d = \left| \overrightarrow{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|,$$

where $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ is normal to the plane.



Example: Find the distance from $S(1, 1, 3)$ to the plane $3x + 2y + 6z = 6$.

Solution: The plane $Ax + By + Cz = D$ has normal vector $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$. If $P(x_0, y_0, z_0)$ is a point on the plane and $S(x_1, y_1, z_1)$ is any point, then

$$\overrightarrow{PS} \cdot \mathbf{n} = (x_1 - x_0)A + (y_1 - y_0)B + (z_1 - z_0)C = Ax_1 + By_1 + Cz_1 - D.$$

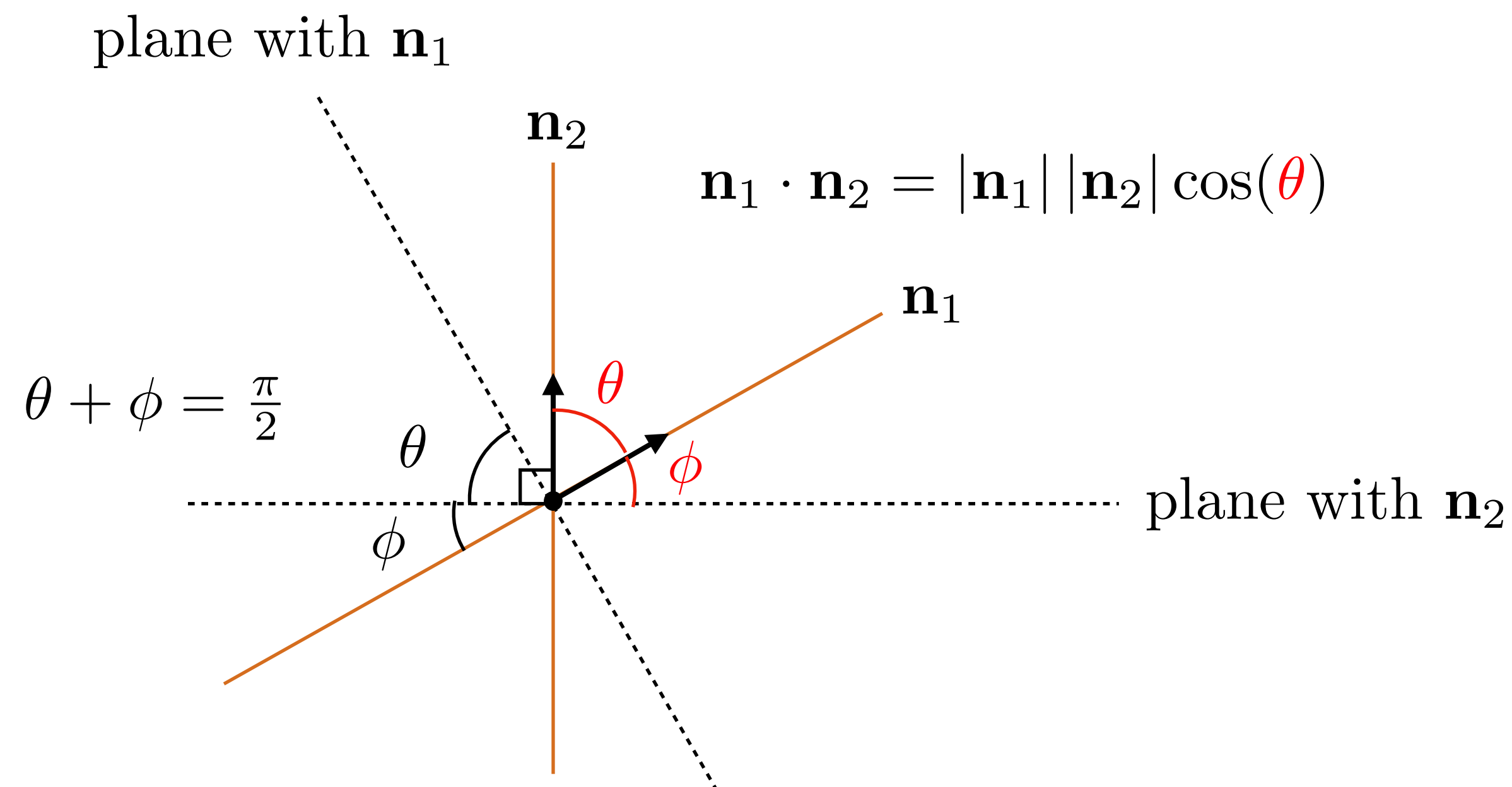
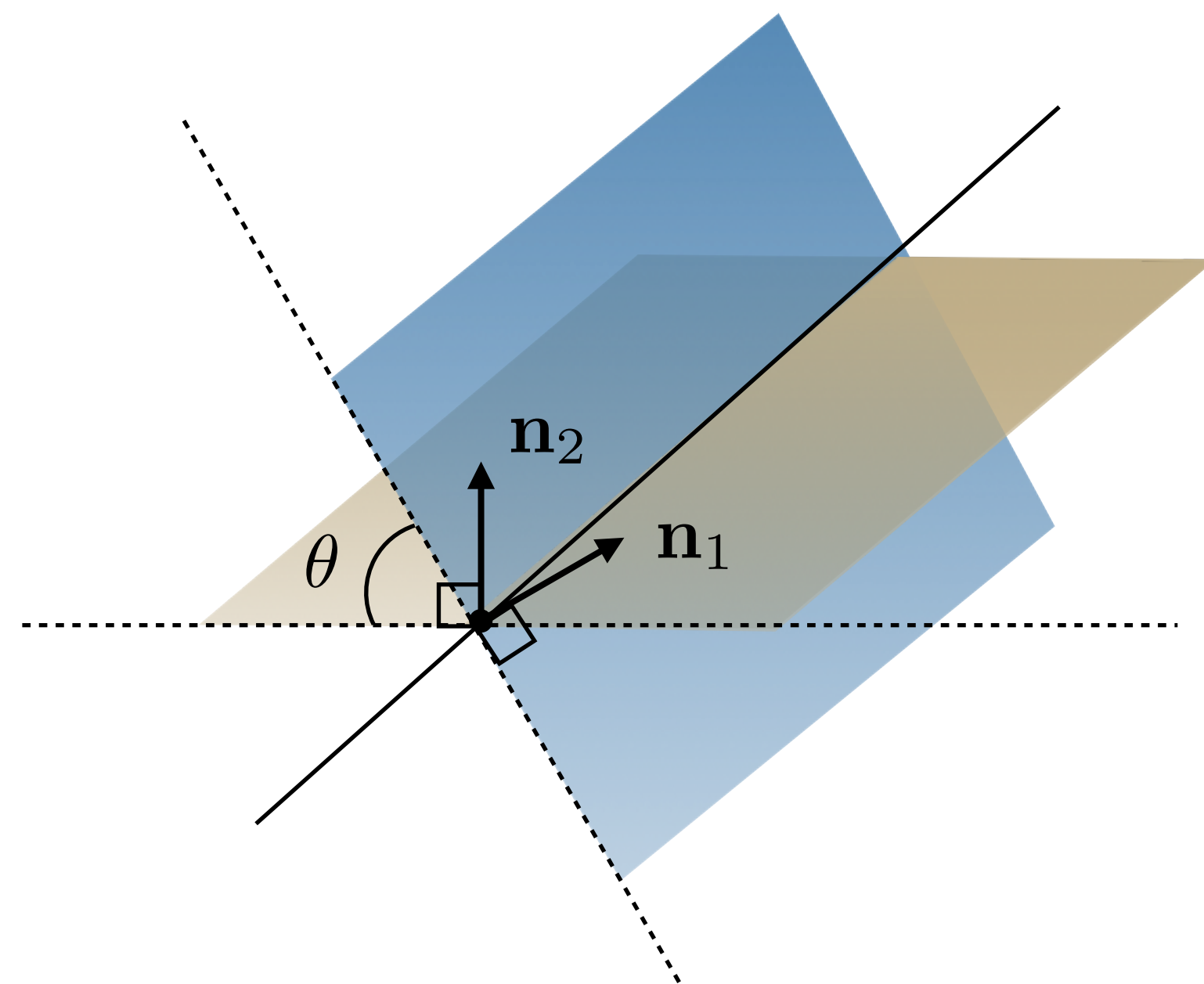
Therefore, the distance from $S(x_1, y_1, z_1)$ to the plane $Ax + By + Cz = D$ is: $d = \frac{|Ax_1 + By_1 + Cz_1 - D|}{\sqrt{A^2 + B^2 + C^2}}$.

Here $A = 3$, $B = 2$, $C = 6$, $D = 6$, and $(x_1, y_1, z_1) = (1, 1, 3)$. Substituting,

$$d = \frac{|3(1) + 2(1) + 6(3) - 6|}{\sqrt{3^2 + 2^2 + 6^2}} = \frac{|3 + 2 + 18 - 6|}{\sqrt{9 + 4 + 36}} = \frac{17}{7}.$$

Angles Between Planes

The angle between two intersecting planes is the acute angle between their normal vectors.



Example: Find the angle between the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

Solution. The normals to the planes can be written in terms of the canonical vectors:

$$\mathbf{n}_1 = 3\mathbf{i} - 6\mathbf{j} - 2\mathbf{k}, \quad \mathbf{n}_2 = 2\mathbf{i} + 1\mathbf{j} - 2\mathbf{k}.$$

The angle θ between the planes is the acute angle between their normals:

$$\cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1| |\mathbf{n}_2|} = \frac{4}{21} \implies \theta = \cos^{-1} \left(\frac{4}{21} \right).$$